Existence of the TV Wasserstein Gradient Flow via JKO scheme

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Math

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Everything You Always Wanted to Know About the JKO Scheme

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Wasserstein Distance

The quadratic Wasserstein distances is defined as:

$$\frac{1}{2}W_2^2(\nu,\mu) = \inf_{\pi}\left\{\int \frac{|x-y|^2}{2}\,\mathrm{d}\pi(x,y) : \int_y \pi(x,y) = \mu, \int_x \pi(x,y) = \nu\right\}$$
$$= \inf_{T}\left\{\int \frac{|x-T(x)|^2}{2}\,\mathrm{d}\mu(x) : T_{\#}\mu = \nu\right\}$$

We recalll the Kantorovich dual formulation of W_2^2:

$$\frac{1}{2}W_2^2(\nu,\mu) = \sup\left\{\int_{\mathbb{R}^d}\psi\,\mathrm{d}\nu + \int_{\mathbb{R}^d}\varphi\,\mathrm{d}\mu : \psi(y)+\varphi(x) \le \frac{|y-x|^2}{2}\right\}$$

an optimal pair (ψ,φ) is called a **Kantorovich potentials**.

Theorem (Brenier). *if $\mu \ll \mathcal{L}^d \implies \varphi$ is differentiable μ a.e., the optimal transport map between μ and ν is given by $T = \mathrm{Id} - \nabla\varphi$.*

Wasserstein Gradient Flow and JKO scheme

Let $\mathcal{G} : \mathbb{P}(\Omega) \to \mathbb{R}$ be a functional. Let $\frac{\delta\mathcal{G}}{\delta\rho}$ be the first variation of $\mathcal{G}$.

The Wasserstein gradient flows $\rho_t : [0,T] \to \mathbb{P}(\Omega)$ is a continuous curve in $\mathbb{P}(\Omega)$ which satiflies the continuity equation:

$$\partial_t\rho_t - \operatorname{div}\left(\rho_t\nabla\frac{\delta\mathcal{G}}{\delta\rho}(\rho_t)\right) = 0 \tag{1}$$

- We can view $-\operatorname{div}\left(\rho\nabla\frac{\delta\mathcal{G}}{\delta\rho}(\rho_t)\right)$ as $\nabla_{W_2}\mathcal{G}(\rho_t)$.

The JKO scheme is the following iterated minimization scheme:

$$\rho_0^\tau = \rho_0, \quad \rho_{k+1}^\tau \in \operatorname{argmin}_\rho\left\{\frac{1}{2\tau}W_2^2(\rho_k^\tau,\rho) + \mathcal{G}(\rho)\right\}$$

Formally, when $\tau \to 0$, the sequence $\{\rho_k^\tau\}$, suitably interpolated, converges to a solution of (1). One can see [1] for more details.

TV Wasserstein Gradient Flow

We consider the gradient flow of the total variation $J(\rho) = \int|\nabla\rho|$ (in the BV sense).

$$\partial_t\rho + \operatorname{div}\left(\rho\nabla\operatorname{div}\left(\frac{\nabla\rho}{|\nabla\rho|}\right)\right) = 0$$

where we viewed the term $-\operatorname{div}\left(\frac{\nabla\rho}{|\nabla\rho|}\right)$ as an element of $\partial J(\rho)$

The JKO scheme of above continuous equation has been studied by Carlier and Poon [2]. But the convergence of the JKO scheme was only presented in 1D.
Why this restriction on the dimension ? some estimates in order to pass the limit $\tau \to 0$ required lower bounds on the density ρ_k^τ. i.e. **a minimun principle at the JKO level is needed.**
Unfortunately when $d \ge 2$, $\|\rho\|_{L^{-p}}^p$ is not a displacement convex functional, we cannot use the flow interchange technique [3] to obtain the lower bound.

Minimun principle preserved in the PDE

If we consider **the minimun principle in the PDE**:
We rewrite the equation as:

$$\partial_t\rho + \operatorname{div}(\rho\nabla\Delta_1\rho) = 0$$
$$\text{where } \Delta_1\rho := \operatorname{div}\left(\frac{\nabla\rho}{|\nabla\rho|}\right) = \nabla\cdot((\nabla G)(\nabla\rho)), \quad G(\mathbf{v}) = |\mathbf{v}|$$

For any H convex, formally:

$$\left.\begin{aligned}
&\frac{d}{dt}\int H(\rho) = \int H'(\rho)\partial_t\rho\\
&= -\int H'(\rho)\nabla\cdot(\rho\nabla\Delta_1\rho)\\
&= \int \rho H''(\rho)\nabla\rho\cdot\nabla\Delta_1\rho\\
&\text{choose } g : g'(s) = sH''(s),\ \text{let } p = g(\rho) \implies \Delta_1\rho = \Delta_1 p\\
&= \int \nabla p\cdot\nabla\Delta_1 p\\
&= \int \sum_i p_i\left(\sum_j (G_j(\nabla p))_{ji}\right)\\
&= -\int D^2p : D^2G : D^2p \le 0
\end{aligned}\right\} \tag{2}$$

which implies that $H(\rho)$ is decreasing along the flow, whenever H is convex.

- **In the PDE:** displacement convexity is not required to obtain the minimal principle.
- **In JKO steps:** displacement convexity is mainly a technical requirement to obtain the minimun principle.
- **Difficulty:** We need a lower bound in the JKO level in order to prove the convergence of the JKO scheme.

Main Proof

Our Idea

- Adding a term $\int f_\epsilon(\rho)$ to give an artificial lower bound at the JKO scheme level.
- At the PDE level, we pass to the limit $\epsilon \to 0$ to get the solution of the TV gradient flow.

We first consider the approximation TV-JKO scheme:

$$\rho_0^{\epsilon,\tau} = \rho_0 \quad \rho_{k+1}^{\epsilon,\tau} \in \operatorname{argmin}\left\{\frac{1}{2\tau}W_2^2(\rho_k^\tau,\rho) + F_\epsilon(\rho)\right\} \tag{3}$$

$$\text{where} \quad F_\epsilon(\rho) = J(\rho) + \int f_\epsilon(\rho), \quad f_\epsilon(s) = \begin{cases}\frac{\epsilon}{s-c} & \text{if } s > c\\ +\infty & \text{otherwise.}\end{cases}$$

- Thanks to the artificial lower bound, the JKO method works. We obtain a weak solution of the approximated gradient flow.

$$\partial_t\rho_\epsilon + \operatorname{div}\left(\rho_\epsilon\nabla\operatorname{div}\left(\frac{\nabla\rho_\epsilon}{|\nabla\rho_\epsilon|}\right) - f'_\epsilon(\rho_\epsilon)\right) = 0$$

- In order to pass $\epsilon \to 0$, we need to build a minimun principle for the approximated gradient flow. We get it by examining what regularity of ρ_ϵ is needed to realize the computations (2):

  Now we view the term $-\Delta_1$as an element of $\partial J(\rho)$, and we we fix $\epsilon > 0$ and omit it for simplicity.

$$\frac{d}{dt}\int H(\rho) = \int H'(\rho)\partial_t\rho \qquad \textbf{The regularity we want:}$$
$$= -\int H'(\rho)\nabla\cdot[\rho\nabla(\Delta_1\rho - f'(\rho))] \qquad \text{Domain } \mathbb{T}^d$$
$$= \int \rho H''(\rho)\nabla\rho\cdot\nabla(\Delta_1\rho - f'(\rho)) \qquad \rho \in L_t^2H_x^1$$
$$\le \int \rho H''(\rho)\nabla\rho\cdot\nabla\Delta_1\rho \qquad f'(\rho) \in L_t^2H_x^1$$

choose $g :\ g'(s) = sH''(s)$ — **The relation needed for** ρ and $\Delta_1\rho$:
let $p = g(\rho) \implies \Delta_1\rho = \Delta_1 p$ — $\Delta_1\rho = \Delta_1 p$

$$= \int \nabla p\cdot\nabla\Delta_1 p \qquad \int \nabla p\cdot\nabla\Delta_1 p \le 0$$

- The regularity of ρ and $f'(\rho)$ can be obtained by the convex duality method [4].

  Theorem. *In one step of the JKO scheme (3), we have $\rho_{k+1}^\tau \in H^1(\mathbb{T}^d)$ with the following estimates:*
  $$\|f'(\rho_k^\tau)\|_{H^1} \le \|-\operatorname{div} z_k^\tau + f'(\rho_k^\tau)\|_{H^1}$$
  Passing the limit $\tau \to 0$, we get $\rho, f'(\rho) \in L_t^2H_x^1$.

  Remark. After we get $\rho_k^\tau \in H^1(\mathbb{T}^d)$, we also can use the flow interchange technique to obtain the regularity of ρ. But that is not enough to get the regularity of $f'(\rho)$.

- $\Delta_1\rho = \Delta_1 g(\rho)$ is easy to get. The inequality can be obtained by using the optimality of $J(\rho) = \int \rho\Delta_1\rho$.

  Theorem. *If $\rho \in H^1(\mathbb{T}^d)$, $\Delta_1\rho \in H^2_{\mathrm{div}}(\mathbb{T}^d)$, then*
  $$\int_{\mathbb{T}} \nabla\rho\cdot\nabla\Delta_1\rho \le 0$$

Main theorem

On the torus $\mathbb{T}^d$, if the initial data $\rho_0 \in \mathbb{P}\cap BV\cap L^\infty$ satisfies $\rho_0 > c > 0$, then the TV Wasserstein gradient flow admits a weak solution which remains bounded from below by c.

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